Engineering Optimization

Unit Commitment Problems In Energy Systems

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1. Introduction

To manage complex infrastructure systems, engineering optimization is necessary. This practice is perhaps nowhere more vital than in the electricity generation industry. System operators must make sure that the power supply precisely matches consumer demand always, a challenge that is complicated by transmission limits, system inertia, and the economic necessity of minimizing operating costs. In this paper, we explore the Unit Commitment (UC) Problem, an optimization method that addresses this challenge.

The UC Problem aims to determine the optimal schedule (when to turn ON or OFF) for all available generating units over a short-term horizon (usually 24 to 168 hours), while in tandem deciding their optimal power output. The main objective is to minimize total possible cost while satisfying the forecasted demand and various operational constraints. Because UC involves determining the binary (ON/OFF) status of generators alongside continuous power output levels, it is classified as a large-scale, non-linear mixed-integer programming problem *[Anjos & Conejo, 2017]*. Methods such as dynamic-programming can solve small-scale UC problems, however; they suffer from the *curse of dimensionality* – the state space grows exponentially with the number of units, making them unsuitable for realistic systems. This paper explores **Lagrange Relaxation Solutions***,* which provide an effective framework to overcome this challenge *[Wood and Wollenberg, 2012].*

2. The Optimization Model

2.1. Decision Variables

To start, define the decision variables.

$$U_{i,t} := \{1 \text{ if unit } i \text{ is online during period } t, \text{ 0 otherwise}\} \ \forall\, i \in N,\ t \in T$$

2.2. Objective Function

The objective is to minimize the total operational cost over the time horizon T.

$$\sum_{t=1}^{T} \sum_{i=1}^{N} [F_i(P_{i,t}) + [startup\ cost]_{i,t}]U_{i,t} = F(P_{i,t}, U_{i,t})$$

where:

$F_i(P_{i,t})$:= the fuel cost function for unit i at power output $P_{i,t}$

N := the number of generating units

T := the number of time periods in the scheduling horizon

$[startup\ cost]_{i,t}$:= the cost to start unit i and time t

2.3. Constraints

This optimization problem is subject to two categories of constraints: system-wide (coupling) constraints and unit-specific (local) constraints.

System Constraints (Coupling):

These constraints link the operation of all units together. Power Balance: the total generation must precisely meet the load demand in each period.

$$\sum_{i=1}^{N} P_{i,t} = P_{load,t} \ \forall\, t \in \{1,\ \dots,\ T\}$$

1. Spinning Reserve: sufficient reserve capacity must be available from online units to handle unforeseen contingencies or load fluctuations.

$$\sum_{i=1}^{N} U_{i,t} P_{i,max} \geq P_{load,\,t} + R_t \ \forall\, t \in \{1,\ \dots,\ T\}$$

where:

$R_t :=$ the minimum system reserve requirement at time t.

Unit-Specific Constraints (Local):

These constraints apply to each unit i individually.

1. Generation Limits: each committed unit must operate within its stable minimum and maximum power output levels.

$$U_{i,t} P_{i,min} \leq P_{i,t} \leq U_{i,t} P_{i,max} \ \forall\, i, t$$

2. Minimum up and down time: once a unit is turned on/off, it must remain on/off for a minimum number of consecutive hours ($T_{i,up}$, $T_{i,down}$)
3. Ramp rate limits: the change in a unit's power output between consecutive periods is limited by its ramp-up (RU_i) and its ramp-down (RD_i) rates.

3. The Lagrange Relaxation Framework

The Lagrange Relaxation method works by moving these difficult coupling constraints *into* the objective function by adding a new variable, a Lagrange multiplier. This process is known as dualization, and creates a new, "relaxed" problem that is easier to solve. *[Wood and Wollenberg, 2012].*

3.1. The Lagrangian Function

The core idea of Lagrange Relaxation is to simplify the problem by moving the difficult, system-wide coupling constraints into the objective function. We achieve this by associate a Langrage multiplier, λ_t, with each loading constraint for each hour t. The multiplier λ_t can be interpreted as the estimated marginal cost of energy (e.g. system price) for that hour.

This creates a new, "relaxed" Lagrangian Function $\mathcal{L}(P, U, \lambda)$.

$$\mathcal{L}(P, U, \lambda) = F(P_{i,t}, U_{i,t}) + \sum_{t=1}^{T} \lambda_t (P_{load,t} - \sum_{i=1}^{N} P_{i,t} U_{i,t})$$

The unit commitment problem requires that we minimize the Lagrange function subject to the local unit constraints (unit limits, unit minimum up and down time constraints). This essentially requires us to rearrange the terms and solve the optimization for each unit independently, breaking the large problem down to N small, manageable subproblems. This decomposition is what allows the Lagrange Relaxation method to efficiently solve large-scale Unit Commitment problems.

3.2. Expanding the Lagrangian Function

To make this easier to solve, we first expand the terms.

$$\mathcal{L}(P, U, \lambda) = \sum_{t=1}^{T} \sum_{i=1}^{N} [F_i(P_{i,t}) + [startup\ cost]_{i,t}]U_{i,t} + \sum_{t=1}^{T} \lambda_t P_{load,t} - \sum_{t=1}^{T} \lambda_t \sum_{i=1}^{N} P_{i,t} U_{i,t}$$

Now, the key insight is to rearrange this equation by grouping all the terms associated with each individual generating unit i.

$$\mathcal{L}(P, U, \lambda) = \sum_{i=1}^{N} (\sum_{t=1}^{T} \{[F_i(P_{i,t}) + [startup\ cost]_{i,t}]U_{i,t} - \lambda_t P_{i,t} U_{i,t}\}) + \sum_{t=1}^{T} \lambda_t P_{load,t}$$

Since we are minimizing $\mathcal{L}$ with respect the unit variables $U_{i,t}$ and $P_{i,t}$, and the multipliers λ_t are currently fixed, the term $\sum_{t=1}^{T} \lambda_t P_{load,t}$ is just a constant. We can therefore drop this constant term without changing the optimal solution. This leaves us with a final, decomposed function:

$$\mathcal{L}(P, U, \lambda) = \sum_{i=1}^{N} (\sum_{t=1}^{T} \{[F_i(P_{i,t}) + [startup\ cost]_{i,t}]U_{i,t} - \lambda_t P_{i,t} U_{i,t}\})$$

We have now successfully separated the problem into N independent pieces, one for each generating unit.

3.3. Solving the Single-Unit Subproblems

To minimize the total function, we must solve N separate, smaller problems. For each unit i, we must solve

$$min \sum_{t=1}^{T} [F_i(P_{i,t}) + [startup\ cost]_{i,t}]U_{i,t} - \lambda_t P_{i,t} U_{i,t}$$

This subproblem is subject to that specific unit's own constraints, such as minimum and maximum power, minimum and maximum up and down times, etcetera.

This single-unit scheduling problem *now* can be solved using Dynamic Programming. The DP algorithm allows a cost-optimal path for the unit through time, deciding whether it should be on or off in each hour t. To make the decision at each hour, the dynamic program must determine two things:

1. Optimal Power Dispatch

If the unit is assumed to be ON ($U_{i,t} = 1$) in some arbitrary hour t, we must find its most economical power output. This is done by minimizing the cost portion of the function for hour t.

$$min_{P_{i,t}} [F_i(P_{i,t}) - \lambda_t P_{i,t}]$$

This minimum is found by taking the derivative with respect to $P_{i,t}$, and setting it to zero.

$$\frac{dF_i(P_{i,t})}{dP_{i,t}} [F_i(P_{i,t}) - \lambda_t P_{i,t}] = 0$$

$$\frac{dF_i(P_{i,t})}{dP_{i,t}} = \lambda_t$$

This is a foundational concept in power systems. A unit is dispatched most economically when its marginal fuel cost, $\frac{dF_i(P_{i,t})}{dP_{i,t}}$, is equal to the system's marginal cost, λ_t.

However, this must be done while respecting the unit's generation limits. This leads to three potential cases.

a) If the calculated optimal dispatch, $P_{i,opt}$, is below the minimum limit, the unit must produce at $P_{i, min}$.
b) If $P_{i,opt}$ is within the limits, the unit produces at $P_{i,opt}$.
c) If $P_{i,opt}$ is above the maximum limit, the unit must produce at $P_{i, max}$.

2. Optimal Commitment (Finding $U_{i,t}$)

The dynamic programming algorithm uses the minimum cost found in part 1 to decide the optimal ON/OFF schedule. A unit is only worth committing if its generating cost is less than the "price" it's being paid, λ_t. So, the unit is economically favorable if:

$$[F_i(P_{i, opt}) - \lambda_t P_{i, opt}] < 0$$

The dynamic programming algorithm evaluates this at each step, while also considering start-up costs and minimum up and down time constraints. This then finds the minimized cost schedule for that single unit over the time period t.

3.5. Adjusting the System's Marginal Cost

After solving the dynamic programming subproblems for all N units, a complete schedule is created. However, it is likely that this schedule is *infeasible*. The total power generation $\Sigma P_{i,t}$ will likely not equal the total load, $P_{i, load}$. So, we are brought back to step 1 of the dualization procedure – updating the λ_t values.

To do this, we implement a subgradient method. The update rule is:

$$\lambda_{t,k+1} = \lambda_{t,k} + \alpha_k g_{t,k}$$
$$g_{t,k} = P_{t,load} - \sum_{i=1}^{N} P_{i,t}(\lambda_k)$$

where:
$\lambda_{t,k}$:= multiplier for hour t at iteration k
α_k:= small positive number, aka the step size
$g_{t,k}$:= the subgradient, which is the power mismatch in hour t

This equation logically makes sense. If generation is too low (e.g. the mismatch is positive), we increase the price λ_t for hour t to encourage more units to turn on or increase their output in the next iteration. If generation is too high (e.g. the mismatch is negative), we decrease the price λ_t for hour t to encourage units to shut down or reduce their output.

Then, this entire process – solving N subproblems then updating the multipliers λ_t – is repeated iteratively until the power mismatches are acceptably minimal. This yields an almost-optimal solution. *[Wood and Wollenberg, 2012].*

4. Applications of Unit Commitment Lagrange Solutions

The Lagrange Relaxation Method has many applications.

1. Minimizing Power Losses and Electricity Costs
One of the biggest applications of the Lagrange Relaxation Framework is applied to the optimal planning and integration of Distributed Generators (DGs) within existing power distribution networks. Modern grids are ever-evolving, and utilities face the complex optimization problem of determining ideal location, capacity, and operational parameters for new DGs that can achieve system-wide objectives such as power loss reduction and electricity cost minimization.

The Lagrange Relaxation method provides the *systemic framework* to solve this problem. It decomposes the large, centralized optimization problem. In this application, the Lagrange multipliers generated during the solution process serve as the "price signals" needed to create effective incentive mechanisms. These multipliers represent the *shadow price* of injecting power at each specific location in the network.

A high multiplier indicates a location where a new DG would be very effective at reducing system losses, while a low multiplier indicates a less ideal location. By translating these mathematical multipliers into tangible financial incentives, such as location-based feed-in tariffs, utilities can guide DG owners away from a hands-off approach, and instead encourage them to install their resources at locations that are most beneficial for the entire system. This maximizes consumer welfare and grid efficiency *[Honarvar Nazari and Ilić, 2010]*.

2. Hydro-Thermal Coordination
The goal of a hydro-thermal coordination system is to create a generation schedule that minimizes total cost (the primary fuel cost of thermal plants) over some given time horizon, thus making it perfect for a Lagrange Relaxation solution.

The challenge is that hydro plants have almost zero fuel cost, but their energy is *not* free. It is limited by the amount of water available in their reservoirs. The decision to use water in one hour affects the amount of water available for all future hours; thus creating a complex optimization problem where the system-wide power demand and the limited water resources must be optimized in tandem at all moments.

To use the Lagrange Relaxation method, we can decompose the two main coupling constraints:
1) Power balance constraint (total generation must meet demand in each hour)
2) Water availability constraint (total water used by a hydro plant over the week cannot exceed its limit)

Relaxing these constraints breaks the large problem into independent subproblems:
1) Thermal subproblems (unit commitment problem for each thermal power plant, solved using dynamic programming)
2) Hydro subproblems (hydro plants need to schedule limited water release over the week to generate the most value, solved via the Lagrange multipliers)

The Lagrange multipliers act as internal prices for electricity and water. These prices coordinate the actions of all generators. For example, the multipliers signal to the hydro plant the most valuable hours to release the limited water supply, most likely during peak demand when the thermal generation would be most expensive. The system then iteratively adjusts these prices until it finds an optimal schedule that respects all constraints *[Bento, Pina, Mariano, Rosario Calado, 2020]*.

5. Sources

http://103.203.175.90:81/fdScript/RootOfEBooks/E%20Book%20collection%20-%202023%20-%20B/EEE/Power-Generation-Operation-Control-Allen-Wood.pdf
[1] A. J. Wood and B. F. Wollenberg, Power Generation, Operation, and Control, 3rd ed. Hoboken, NJ, USA: John Wiley & Sons, 2012.

https://www.nowpublishers.com/article/DownloadSummary/EES-014
[2] M. F. Anjos and A. J. Conejo, Unit Commitment in Electric Energy Systems. Hanover, MA, USA: Now Publishers Inc., 2017.

https://ieeexplore.ieee.org/stamp/stamp.jsp?tp=&arnumber=5589994

[3] A. Tuohy, P. Meibom, E. Denny, and M. O'Malley, "Unit Commitment With Wind and Pumped-Storage Hydro," IEEE Transactions on Power Systems, vol. 26, no. 2, May 2011.

https://media.neliti.com/media/publications/510003-short-term-hydro-thermal-coordination-by-be70ffc3.pdf
[4] P. M. S. Bento, A. P. Pina, S. J. P. S. Mariano, and M. do Rosário Calado, "Short-Term Hydro-Thermal Coordination by Benders Decomposition," Energies, vol. 13, no. 22, Nov. 2020.